

AI for Healthcare

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DermaVision AI

Explainable Skin Lesion Classification and Segmentation

Abstract

I wanted this project to do more than give one disease label for an image. A classification model can tell what a lesion resembles, but it does not show where the lesion is. A segmentation model can outline the affected region, but it does not provide a complete image-level prediction. DermaVision AI combines both tasks in one pipeline: it predicts the lesion category, produces a pixel-level lesion mask, calculates the approximate image area covered by the lesion, and presents the outputs together.

The models were trained using transfer learning on 9,547 dermoscopic images from seven lesion categories. The classification model achieved 83.04% test accuracy and a macro F1-score of 66.75%. The segmentation model achieved a test mask mAP50 of 49.96% and mask mAP50-95 of 39.92%. The project is an academic image-analysis prototype and not a medical diagnostic system.

1. Problem Statement

Different skin lesions can look similar in colour, texture, border and overall shape. This makes automated analysis difficult, especially when the model has access only to an image and no patient history. A classification-only system gives a category but does not explain the spatial region used for the prediction. A segmentation-only system gives the lesion boundary but not the final category. The problem addressed in this project was therefore to build one interpretable workflow that answers both questions: what type of lesion is present, and where is it located in the image?

The proposed solution uses two pretrained YOLOv8 nano models. One model performs seven-class image classification and the other performs instance segmentation using polygon annotations. Their outputs are combined to generate a predicted category, confidence score, lesion overlay and estimated lesion-area percentage.

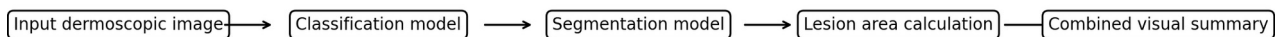


Figure 1. End-to-end DermaVision AI workflow.

2. Dataset and Data Preparation

The dataset was downloaded from the MakhResearch Skin Lesion Segmentation and Classification repository on Hugging Face. It contains 6,675 training images, 1,911 validation images and 961 test images, giving a total of 9,547 images. Each image has a YOLO polygon annotation that stores the lesion class and normalized boundary coordinates.

Code	Lesion category	Train	Validation	Test
BKL	Benign keratosis-like lesion	217	73	36
NV	Melanocytic nevus	361	98	55
DF	Dermatofibroma	446	126	64
MEL	Melanoma	82	21	12
VASC	Vascular lesion	782	216	115
BCC	Basal cell carcinoma	4681	1356	664
AKIEC	Actinic keratosis / intraepithelial carcinoma	106	21	15

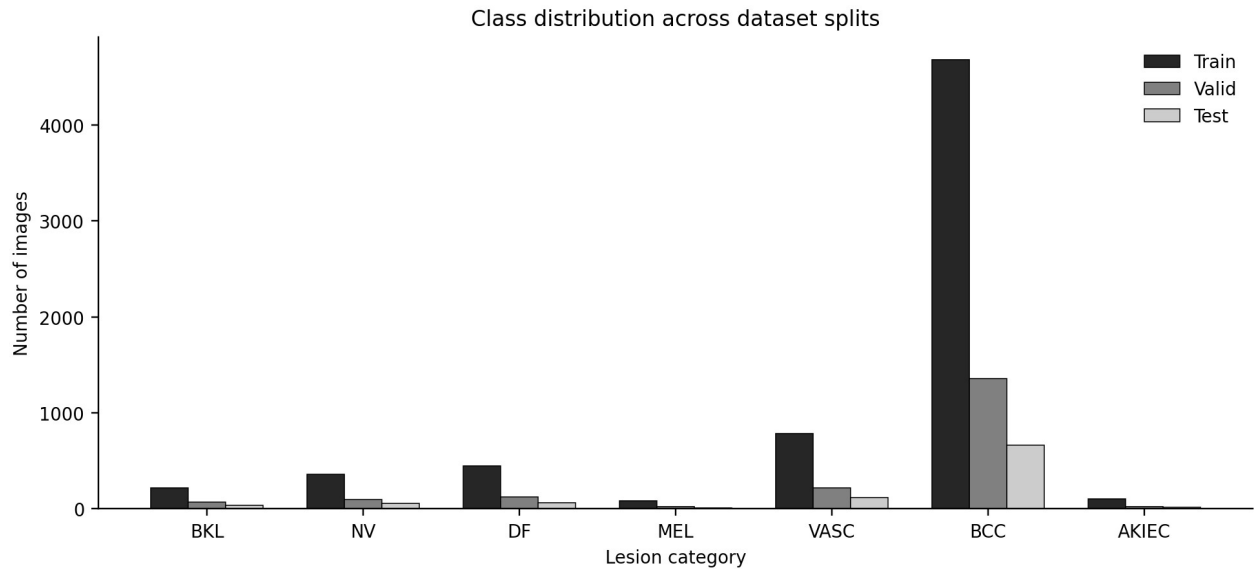


Figure 2. The dataset is highly imbalanced, with BCC forming the majority class.

The imbalance was substantial: the largest training class, BCC, contained 4681 images, while the smallest class, MEL, contained 82. The majority-to-minority ratio was 57.09:1. This is why overall accuracy alone was not treated as sufficient; macro F1-score, per-class recall and the confusion matrix were also used.

Before training, the notebook checked that the train, validation and test folders were present, verified image-label filename matching, searched for empty or malformed annotations, and confirmed that class IDs were valid. A sample polygon was also converted from normalized coordinates to pixel coordinates and drawn over its image. This visual check helped confirm that the label and lesion boundary were aligned.

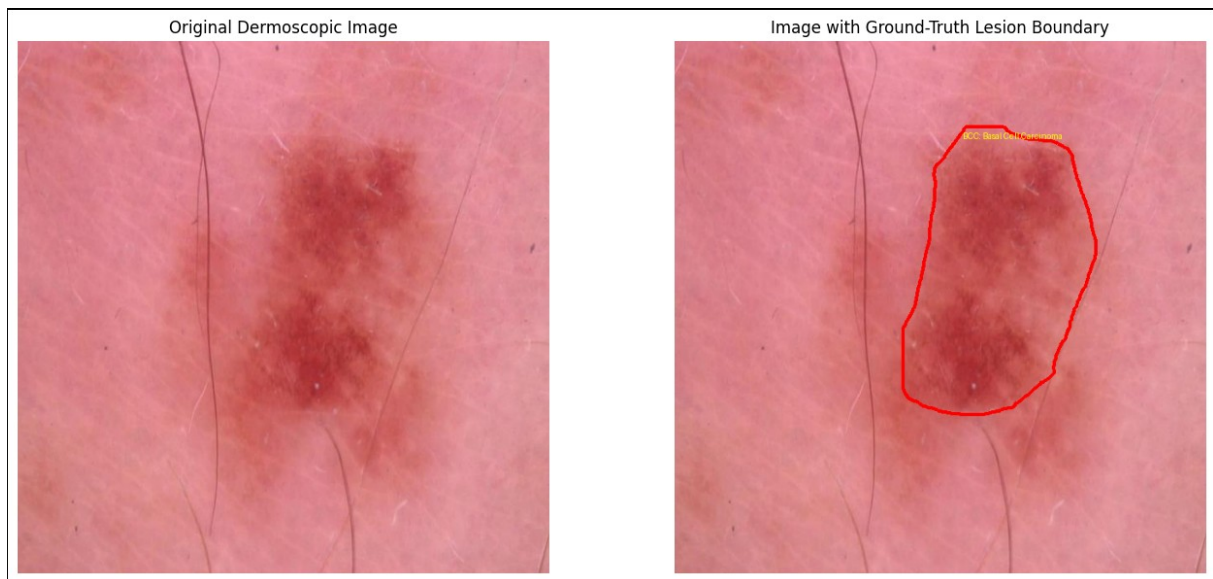


Figure 3. Example of a dermoscopic image and its ground-truth polygon annotation.

3. Methodology

3.1 Segmentation model

For segmentation, I used the pretrained YOLOv8n-seg model. The nano variant was chosen because the complete project had to be trained in a limited Colab session while still leaving enough time for classification, evaluation and integration. Transfer learning allowed the model to start with general visual features such as edges, textures and shapes instead of learning everything from random initialization.

The segmentation model was fine-tuned for 8 epochs at an input size of 416 x 416 pixels, with batch size 16. The dataset YAML file supplied the training, validation and test image paths and the ordered class mapping. The best validation checkpoint, rather than simply the final epoch, was selected for test evaluation.

3.2 Classification model

The same source images were rearranged into class-specific folders for image classification. This avoided using a second dataset with a different visual domain. A pretrained YOLOv8n-cls model was fine-tuned for 8 epochs at 224 x 224 pixels with batch size 64. The classifier produced a probability distribution across BKL, NV, DF, MEL, VASC, BCC and AKIEC.

3.3 Integrated inference

For each new image, the classifier first generated the top predicted class, confidence and top three probabilities. The segmentation model then produced one or more lesion masks. All masks were combined into a binary union mask and resized back to the original image dimensions. The lesion-area percentage was calculated as:

$$\text{Lesion area (\%)} = (\text{predicted lesion pixels} / \text{total image pixels}) \times 100$$

This percentage is only a geometric measurement inside the image. It is not a clinical severity score or disease stage.

4. Results

4.1 Segmentation results

Metric	Test result
Box Precision	53.69%
Box Recall	52.52%
Box mAP50	50.34%
Box mAP50-95	40.90%
Mask Precision	53.45%
Mask Recall	52.10%
Mask mAP50	49.96%
Mask mAP50-95	39.92%

The most relevant metrics for this project are the mask scores. The model reached 49.96% mask mAP50 and 39.92% mask mAP50-95. Mask precision was 53.45% and mask recall was 52.10%. This shows that the model learned meaningful lesion regions, but boundary quality and missed or extra regions remain important limitations.

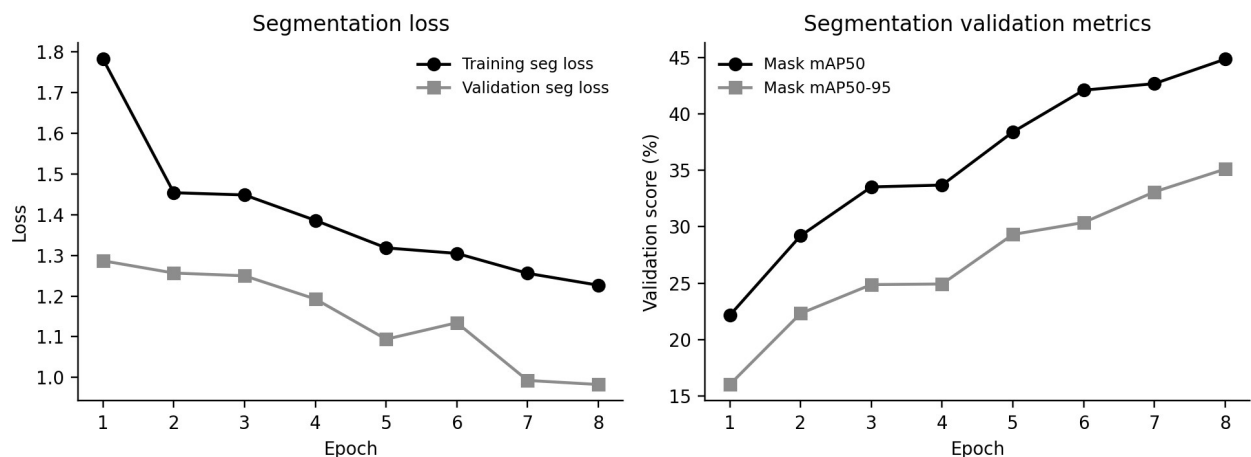


Figure 4. Segmentation loss and validation mask metrics across 8 epochs.

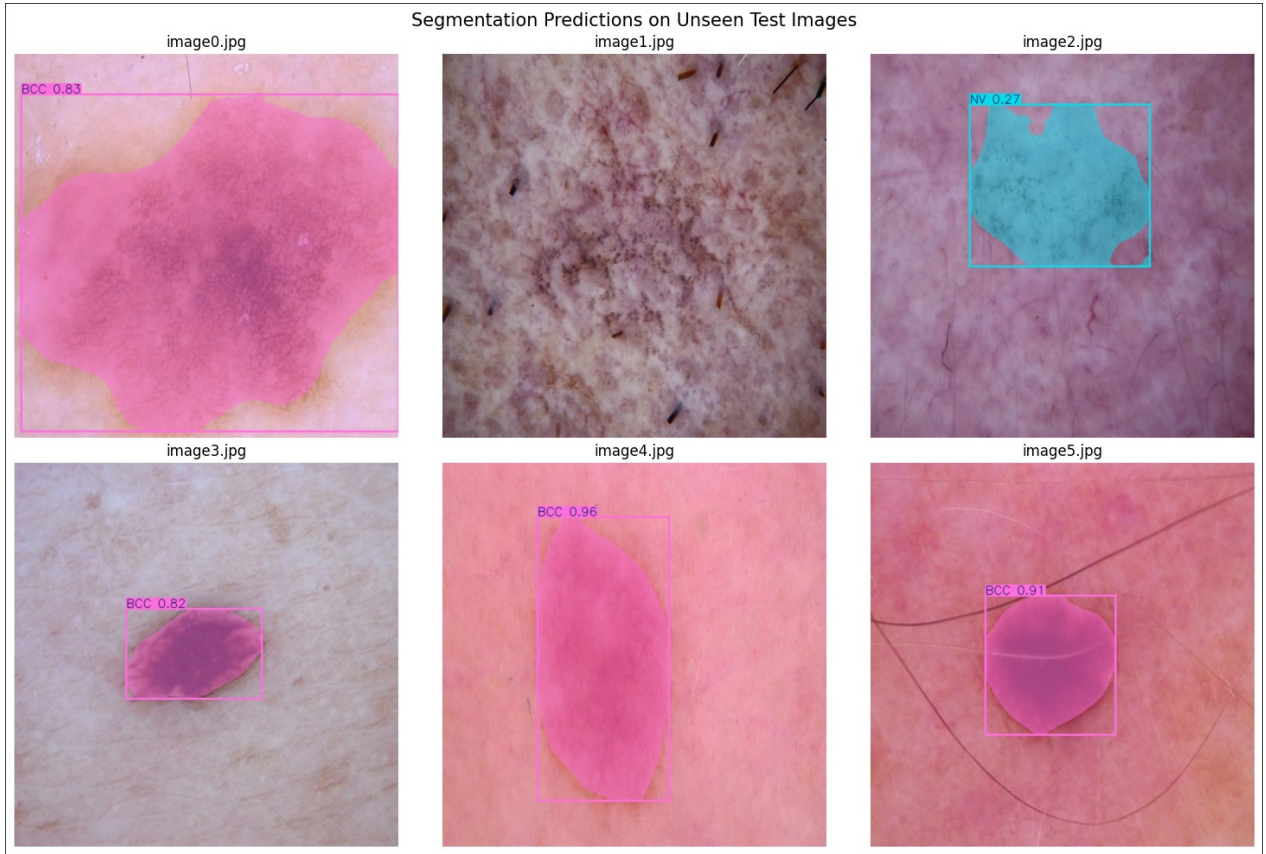


Figure 5. Segmentation predictions on unseen test images. Some lesions are captured well, while some images receive weak or missing predictions.

4.2 Classification results

The classifier correctly predicted 798 of 961 test images, giving an accuracy of 83.04%. The macro F1-score was 66.75% and the weighted F1-score was 82.61%. The gap between macro and weighted F1 reflects the strong class imbalance: performance on the dominant BCC class contributes heavily to the weighted score.

Class	Precision	Recall	F1-score	Support
BKL	67.7%	58.3%	62.7%	36
NV	74.1%	78.2%	76.1%	55
DF	56.1%	50.0%	52.9%	64
MEL	66.7%	33.3%	44.4%	12
VASC	58.6%	56.5%	57.5%	115
BCC	90.7%	93.7%	92.1%	664
AKIEC	91.7%	73.3%	81.5%	15

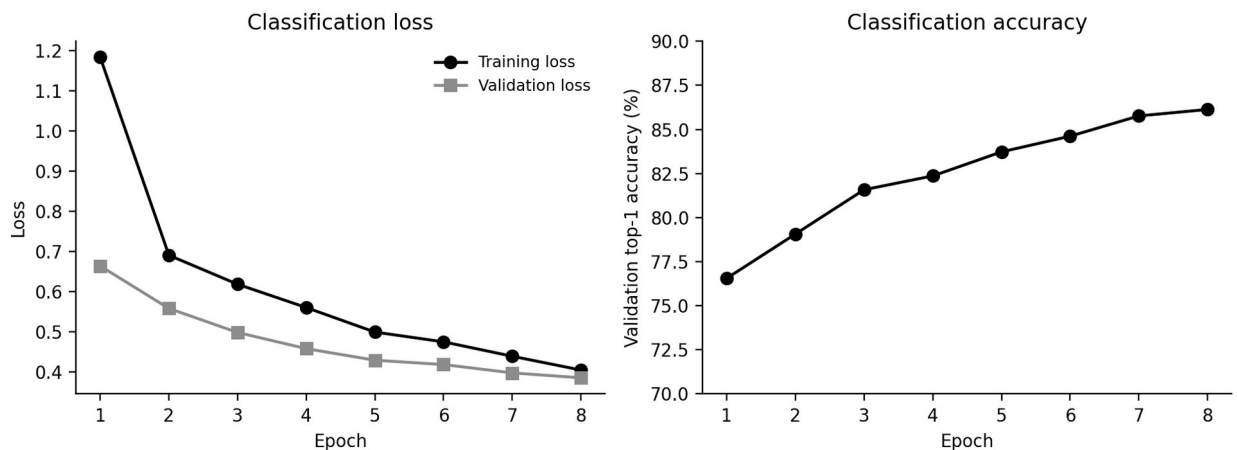


Figure 6. Classification training and validation performance.

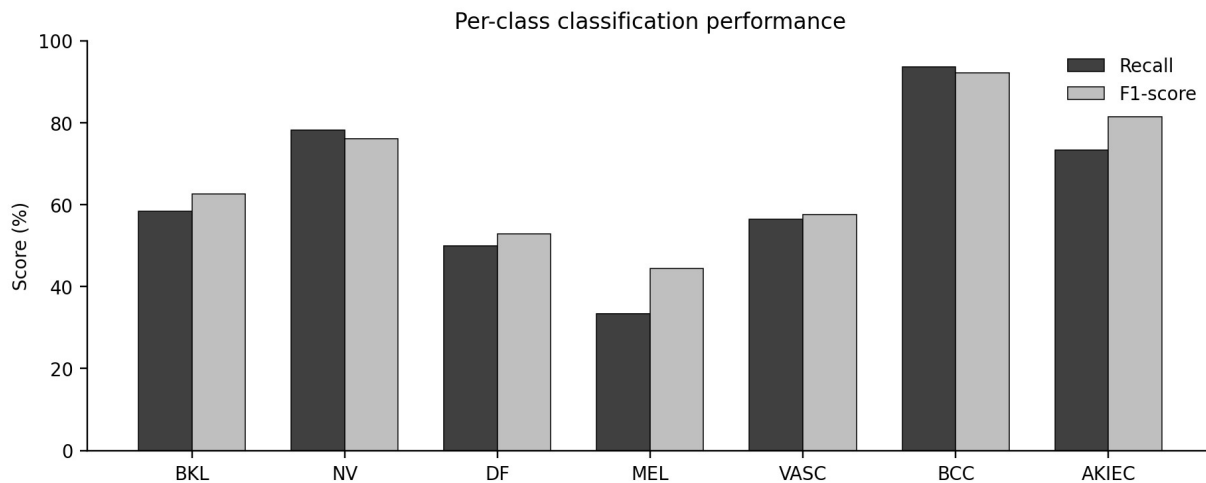


Figure 7. Per-class recall and F1-score show weaker performance on minority classes.

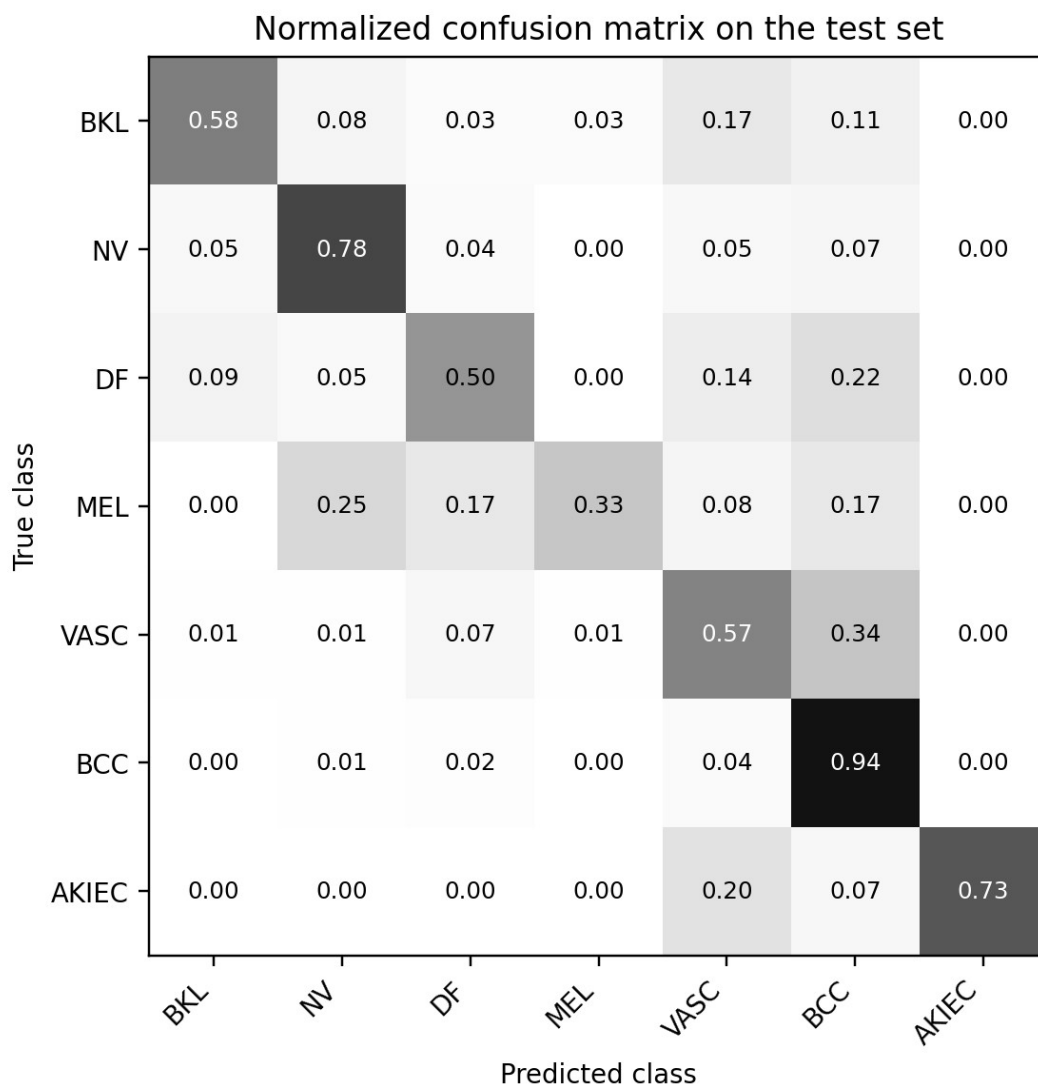


Figure 8. Normalized test confusion matrix.

BCC had the strongest result, with 93.67% recall and 92.15% F1-score, which is consistent with its much larger training set. MEL had only 12 test images and reached 33.33% recall. The most common confusion was VASC predicted as BCC (39 cases), followed by BCC predicted as VASC (24 cases). These errors show why the model should not be judged using overall accuracy alone.

4.3 Integrated pipeline demonstration

Image	Class	Confidence	Lesion area	Masks
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ISIC_0032979.jpg	BCC	45.72%	3.87%	1
ISIC_0030441.jpg	BCC	99.97%	15.29%	1
ISIC_0024563.jpg	BCC	97.83%	26.07%	1

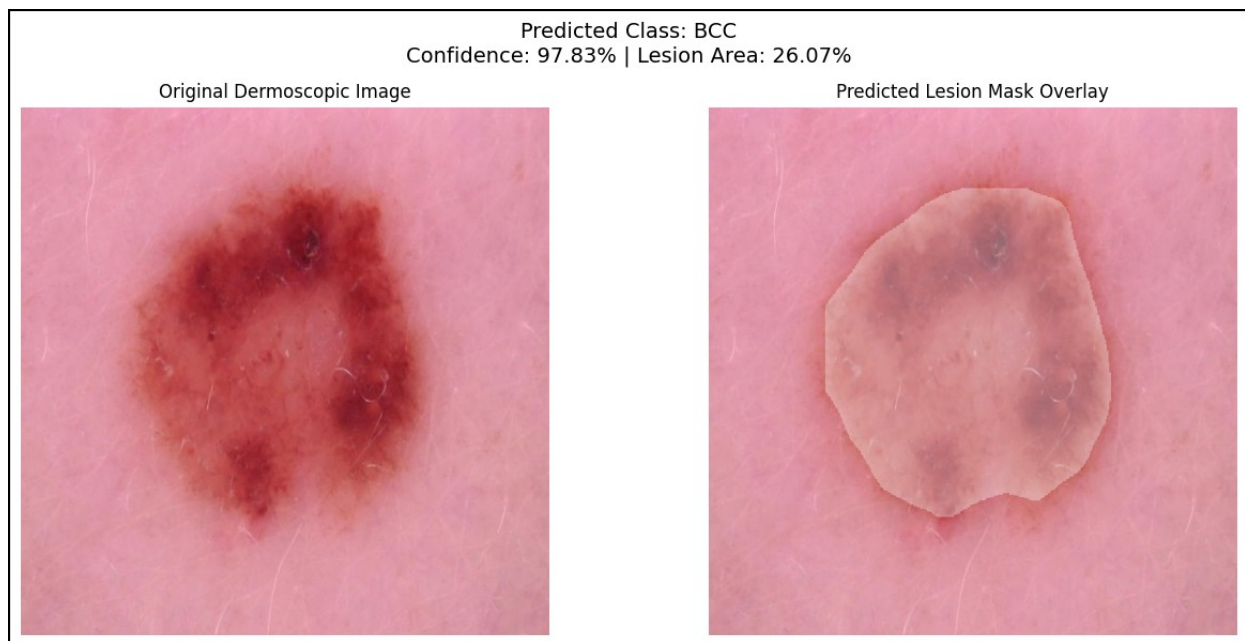


Figure 9. Integrated output with a high-confidence BCC prediction and a lesion mask covering 26.07% of the image.

The integrated pipeline successfully combined the two independently trained models into one readable output. It reported the predicted category and confidence, generated a lesion overlay, and returned an area estimate. This makes the result more interpretable than a classification label alone.

5. Failure Analysis

The project also examined cases where the output looked confident but was not reliable. In the example below, the classifier predicted BCC with only 45.72% confidence, while the segmentation model highlighted a small secondary region instead of the main lesion. This is a useful reminder that a polished overlay can still hide an upstream model error.

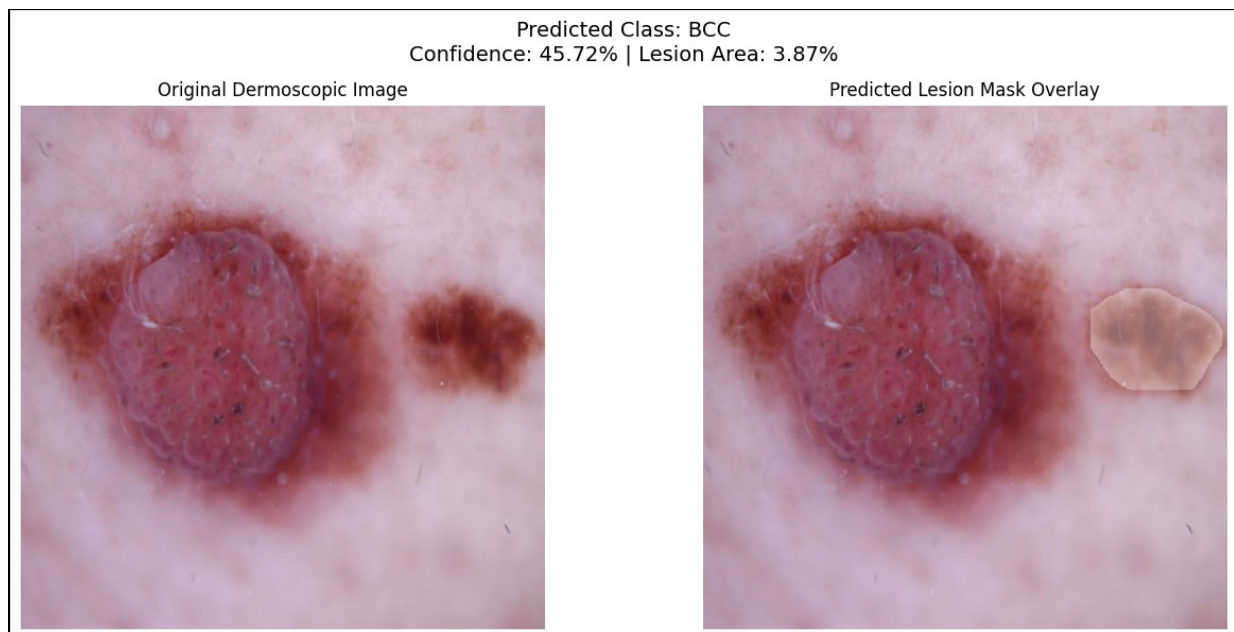


Figure 10. Integrated failure case: the segmentation model focuses on a small secondary region.

High-confidence classification errors were also inspected. Several vascular-lesion images were predicted as BCC with confidence above 97%. These are more concerning than uncertain errors because the model appears

certain while being wrong. Such cases can arise from visual similarity, class imbalance and insufficient representation of minority patterns.

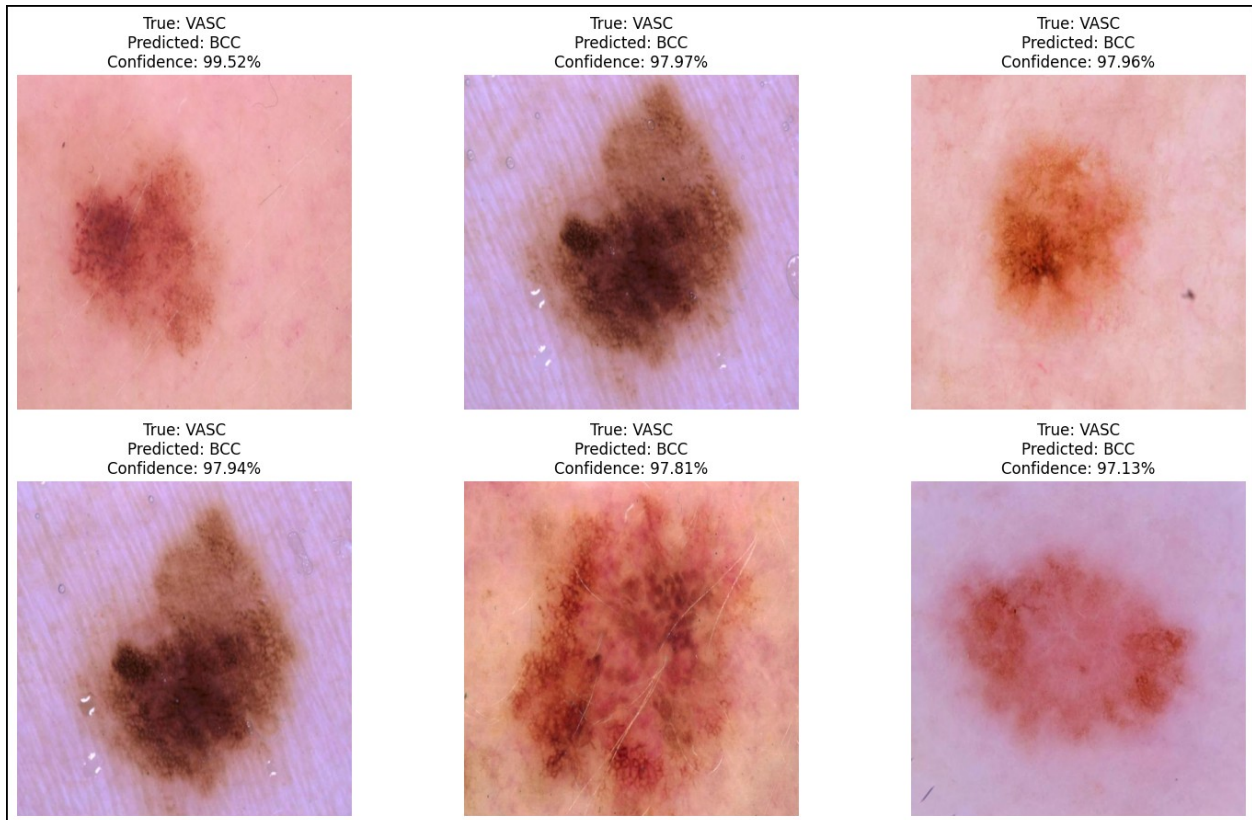


Figure 11. High-confidence VASC-to-BCC classification errors.

6. Limitations and Future Work

- The dataset is strongly imbalanced, and minority classes such as MEL and AKIEC have very few examples.
- Both models were trained for only 8 epochs due to the available time and compute budget.
- Training and testing used splits from the same source dataset; performance on a different clinic or camera is unknown.
- The system uses only the image and does not include symptoms, lesion history, patient age or dermatologist observations.
- The lesion-area percentage depends directly on segmentation quality and must not be interpreted as disease severity.
- The prototype has not undergone external clinical validation, calibration, fairness testing or regulatory review.

The next improvements should focus on class-balanced sampling or class-weighted loss, longer training, comparison with a larger backbone, probability calibration and evaluation on an external dataset. Grad-CAM or another attribution method could be added to explain the classification decision. A carefully labelled Streamlit or Gradio interface could then be created as a research demonstration, not as a diagnostic application.

7. Conclusion

DermaVision AI demonstrates a complete healthcare computer-vision workflow rather than a single model call. The project includes dataset validation, annotation inspection, class-imbalance analysis, transfer-learning-based segmentation, seven-class image classification, test-set evaluation, failure analysis and a combined inference function.

The classifier achieved 83.04% test accuracy with a macro F1-score of 66.75%, while the segmentation model achieved 49.96% mask mAP50. The final pipeline can produce a category, confidence score, lesion overlay and approximate area measurement from one image. The results are promising as an academic prototype, but the error analysis clearly shows why the model should not be treated as a medical diagnosis system.

References

- [1] MakhResearch. Skin Lesion Segmentation and Classification Dataset. Hugging Face Datasets.
- [2] Ultralytics. YOLOv8 Documentation: Classification and Instance Segmentation.
- [3] Project notebook and model outputs generated in Google Colab using PyTorch and Ultralytics.